2025 Data Center Power Report

From Gridlock To Growth: How Power Bottlenecks Are Reshaping Data Center Strategies

January 2025

Executive Summary

The fourth industrial revolution is underway, fueled by rapid advancements in AI and cloud computing. In the US, the rapid deployment of new data center capacity is a strategic priority, but there is a major bottleneck: power availability. Demand for power is only growing, while the electricity grid is aging and new grid projects face permitting and supply chain challenges.

While these challenges have received significant attention, there has been comparatively less discussion about potential solutions. Bloom Energy, a leader in power solutions, explains in this **2025 Data Center Power Report** how data center leaders are shifting paradigms and adopting innovative solutions to meet their strategic goals and economic imperatives.

In April and November 2024, we surveyed data center leaders directly involved in making power architecture decisions. Survey respondents represented a variety of hyperscaler and colocation developer companies and ranged from managers to C-suite executives.[1] To supplement findings from these quantitative surveys, we also commissioned in-depth qualitative interviews with decision makers for data center power architecture and referenced multiple external data sources, including research by the Lawrence Berkeley National Laboratory, McKinsey & Company, and Goldman Sachs.

Summary of findings

1. **Data center leaders expect power availability to get worse**

   Demand for power in the US is growing at an unprecedented rate, with data centers being the largest driver of this growth. The US grid has not been able to keep pace with this demand, and new data center projects will struggle to get timely access to power.

2. **Leaders are increasingly turning to onsite power as a solution**

   Data center leaders expect approximately 30% of all data center sites to use some onsite power as a primary energy source supplemental to the grid by 2030, 2.3 times more than just seven months prior. We find that new data center announcements corroborate this expectation.

3. **Leaders see value differently as they balance priorities**

   New data centers are balancing more priorities, and time to power is playing an increasingly important role in the value equation. Our surveys and interviews with data center leaders have surfaced seven key criteria for choosing onsite power technology.

4. **Looking ahead, leaders expect a paradigm shift in 2025**

   Based on what we have heard from data center leaders, we have identified three key trends we expect to continue in 2025.

1 Surveys were commissioned to enable a double-blind process between Bloom Energy and survey respondents

1. Data center leaders expect power availability to get worse

Demand for power in the US is growing at an unprecedented rate

After 20 years of flat demand,[2] US power needs are predicted to rise by 83 terawatt-hours (TWh) in 2025[3] – equivalent to powering an additional 7.7 million homes.[4]

Data centers are the largest driver of this growth

The US is expected to see the highest share of new data centers outside China. Since 2020, the US colocation data center market alone has doubled, driven by digitization, cloud, and AI.[5]

To power this, data centers by 2030 could require 8-12% of the total US demand compared to 3-4% today.[6]

The US grid has not been able to keep pace with this demand

While utilities can likely generate sufficient power to meet data center needs, they face bottlenecks with transporting that power via **transmission** and **distribution** infrastructure. As a result, grid interconnection takes longer, there is more congestion on the network, and capacity is increasingly expensive.

If the US continues to build high-voltage transmission infrastructure at its current rate, it will take at least 80 years to deliver what we need over the next decade.[7]

New data center projects will struggle to get timely access to power

In the US, 55 GW of data center IT capacity is expected to come online in the next five years. For comparison, this is 10 times the average power capacity used by New York City[8] – and does not include the additional power needed for cooling systems.

We are already seeing data center IT capacity buildout ramp up with ~20 GW of capacity announced so far, and we expect it to continue growing.

We expect that at least another 35 GW of data center capacity will be announced within the next five years to meet projected data center demand.

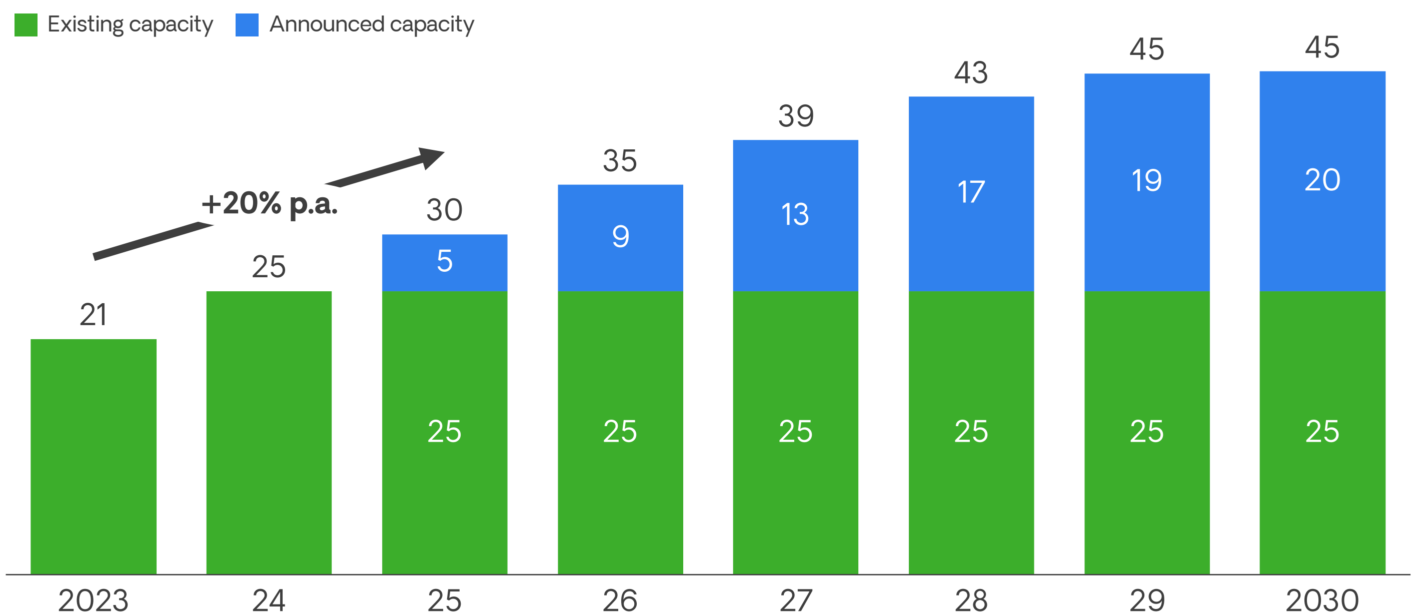
Transmission:
High-voltage transportation of energy across longer distances, from power generation sources to substations

Distribution:
Lower-voltage transportation of energy across shorter distances, from substations to end customers

2 Grid Strategies, "Strategic Industries Surging: Driving US Power Demand"
3 International Energy Agency, "Electricity Mid-Year Update"
4 U.S. Energy Information Administration, "How much electricity does an American home use?"
5 Forbes, "Data Centers At The Heart Of The AI And Digital Economy Surge"
6 Lawrence Berkeley National Laboratory, "2024 United States Data Center Energy"; Goldman Sachs, "Generational growth: AI, data centers and the coming US power demand surge"; McKinsey & Company, "How data centers and the energy sector can sate AI's hunger for power"
7 U.S. Department of Energy, "National Transmission Needs Study"; Americans for a Clean Energy Grid, "Transmission Projects Ready to Go: Plugging Into America's Untapped Renewable Resources"; C Three Group via Robert Bryce, "Out of Transmission"
8 NYC Mayor's Office of Climate & Environmental Justice, "Systems"

20 GW of upcoming data center IT capacity has been announced in the US

GW of cumulative data center IT capacity in the US

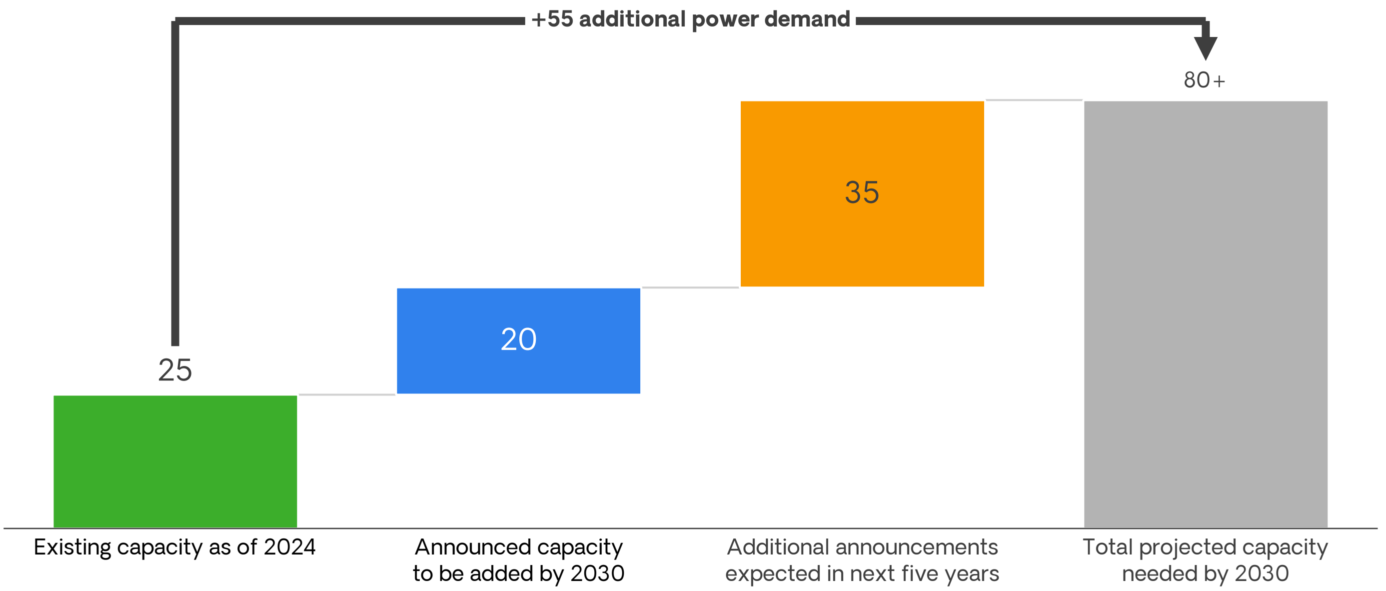


Source: Publicly available data center announcements; McKinsey & Company, "How data centers and the energy sector can sate AI's hunger for power"; Reuters, "US electric utilities brace for surge in power demand from data centers"

---

35 GW of data center IT capacity announcements are expected in the next five years

GW of data center IT capacity in the US



Source: Publicly available data center announcements; McKinsey & Company, "How data centers and the energy sector can sate AI's hunger for power"; Reuters, "US electric utilities brace for surge in power demand from data centers"

2. Leaders are increasingly turning to onsite power as a solution

The pace of change is accelerating

Our research sheds light on what data center leaders are thinking and doing to solve **time to power** challenges. In hearing from these leaders, we were surprised by the pace of change and the growing expectation that **onsite power** generation will play a greater role in powering data center projects.

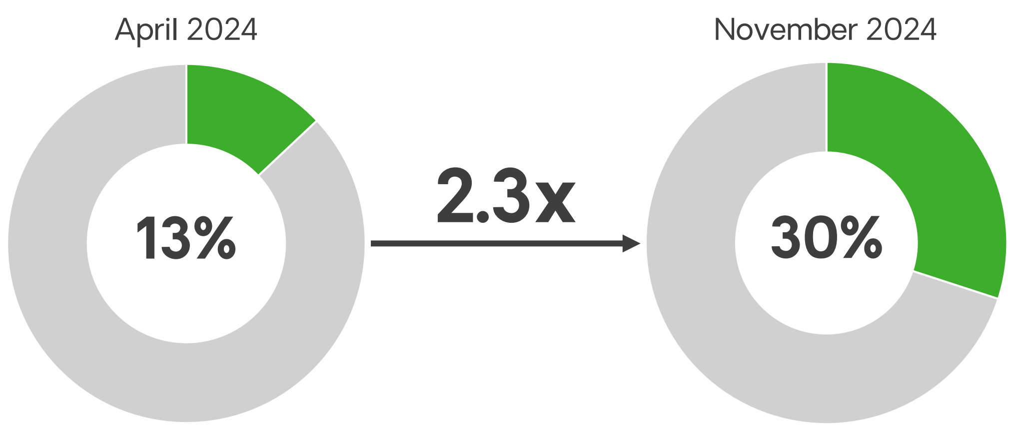
More data center leaders are taking responsibility for their power needs

They do so by generating power onsite, working in close partnership with utilities to lessen the burden on the collective power grid and transmission infrastructure. Historically, data centers have used onsite power mostly for backup purposes, but we are seeing a shift toward onsite power generation as a primary source of power.

Data center leaders increasingly expect onsite power to be adopted as a primary energy source

Leaders expect approximately 30% of all data center sites to use some onsite power by 2030, 2.3 times more than just seven months prior. We find that new data center announcements corroborate this expectation. In 2024, there were more announcements featuring onsite power than the previous four years combined. Total announced capacity of onsite power is now up to 8.7 GW globally, including 4.8 GW expected before 2030.[9]

Average share of data center sites expected to have at least some onsite power generation in 2030, as reported by data center leaders[10]



Source: Survey of data center power architecture decision makers (n=30 in Apr 2024, n=64 in Nov 2024)

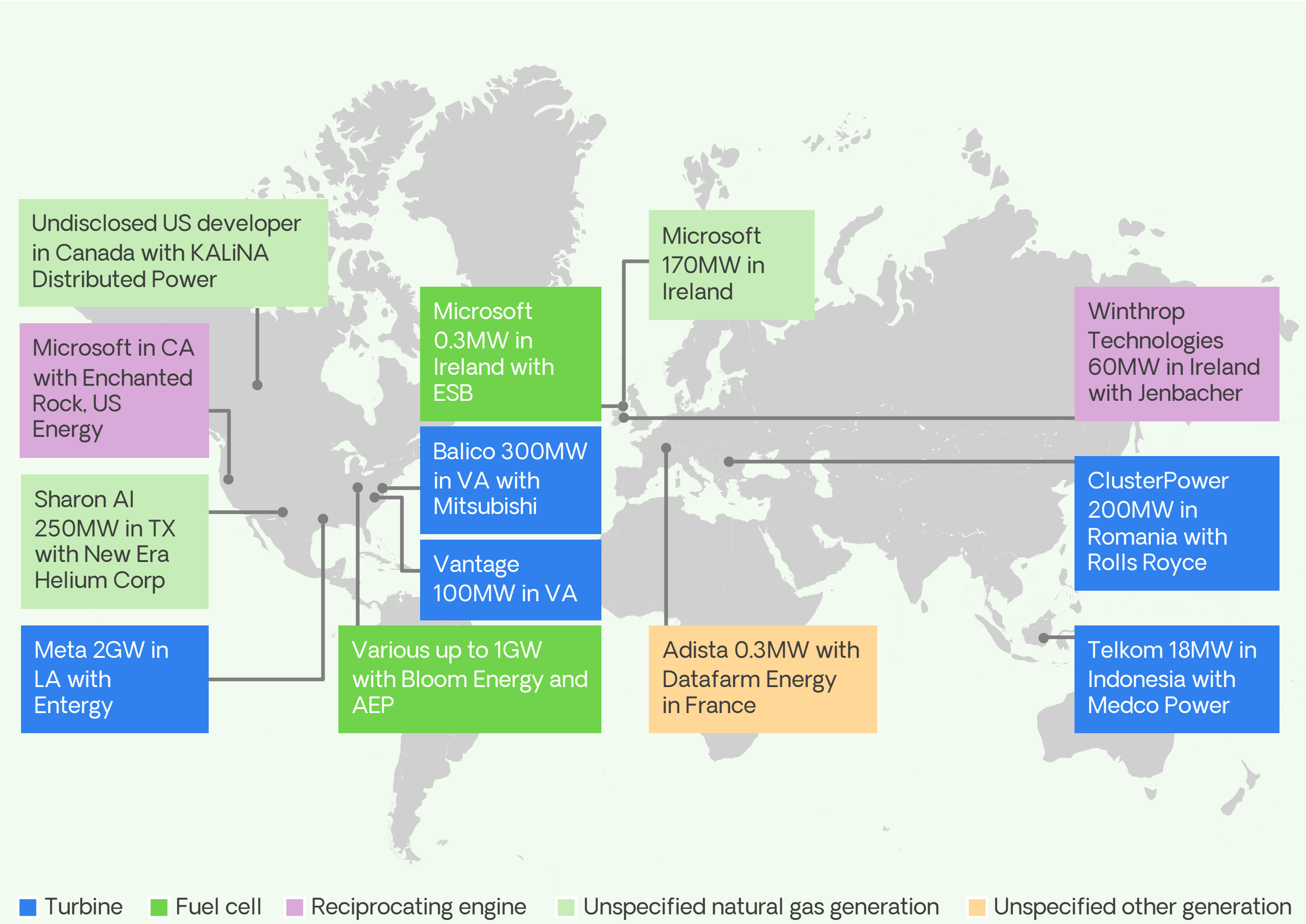
9 Based on publicly available announcements – may not be comprehensive
10 Survey question: What percentage of data center sites do you think will have at least some onsite power generation in 2030?

Time to power:
Time it takes for a new facility to receive necessary power

Onsite power:
Onsite power (also referred to as decentralized or distributed power) means generating electricity directly at the location where it will be used, rather than relying solely on the grid. It can work alongside grid power or operate independently in "islanded mode." Common onsite power technologies include solar panels, small gas turbines, fuel cells, and diesel generators. Generating power directly at the source of consumption helps avoid lead times in transmission infrastructure availability, enabling data centers to come online and begin generating revenue sooner

Data center sites around the world are adopting onsite power solutions

Recently announced data centers using onsite power



Source: Publicly available data center announcements

3. Leaders see value differently as they balance priorities

New data centers need to balance more priorities

Through our surveys and interviews, we asked data center leaders what matters when they consider different technologies and solutions. We learned that decision makers use seven primary criteria to evaluate options.

Notably, it is no longer just a cost and reliability question. Instead, they also factor in the value and impact of time to power, sustainability, power density, and in some cases, the ability to support more demanding and fluctuating AI workloads.

Time to power plays a key role in the value equation

We found that leaders are now ready to invest 50% more than seven months ago if that means they can access power faster for upcoming data center projects.[11]

This shift underscores an important insight: being online ahead of competitors is a strategic advantage, particularly for AI data centers. Leaders recognize that accelerating power access is key to securing a leading position in this rapidly evolving landscape.

Seven key criteria for choosing onsite power technology

1. Reliability

Minimizing outages with reliability at par or superior to the grid (at least 99.9% and sometimes up to 99.9999%) is critical to delivering on the business and customer needs of a data center

2. Time to power

Bringing a data center online faster can provide significant economic advantages (e.g., earlier revenue opportunities) and strategic advantages (e.g., leadership in AI)

3. Cost

Beyond the cost of a particular technology, leaders consider value and return on investment more holistically, factoring in both economic benefits and strategic benefits

4. Load flexibility

AI data centers involve greater power demands and fluctuations over short time frames, and not all power solutions can handle these load shifts

5. Sustainability: Air quality and noise

There is increasing scrutiny of data centers' impact on their local communities and a need to comply with local permitting standards

6. Sustainability: Emissions intensity

Data centers must comply with companies' sustainability commitments as well as carbon emissions regulations

7. Power density

Technologies with greater power density enable a more efficient use of space for revenue-generating IT equipment

11 Based on a six-month acceleration in time to power compared to the grid

Unpacking onsite power technologies

Recent announcements feature a range of solutions as data center players balance the above criteria in the context of each project. Some examples include:

Gas turbines

Combined-cycle gas turbines (CCGT) use a gas turbine and a steam turbine to generate electricity. They are primarily seen in large AI training data centers that require 500 MW or more capacity, as they are cost effective at this scale. Smaller inferencing data centers – focused on the application of trained models – often rely on simple cycle turbines, sometimes paired with batteries. While these technologies are currently the most common onsite power solution, long supply chain lead times and higher carbon dioxide emissions are limiting factors.

Fuel cells

Fuel cells provide continuous electricity by converting natural gas or hydrogen into electricity. They are gaining popularity in data center projects because they are quick to deploy and produce fewer emissions and less pollution than simple cycle turbines.

While fuel cells can be more expensive than some other solutions, data center leaders often prioritize them for their faster deployment, easier permitting, and greater power density, which deliver both economic and strategic advantages. Since fuel cells generate direct current (DC) electricity, they are also poised to continue meeting power demands as data centers increasingly explore DC for the reliability and efficiency benefits of its simplified power architecture.

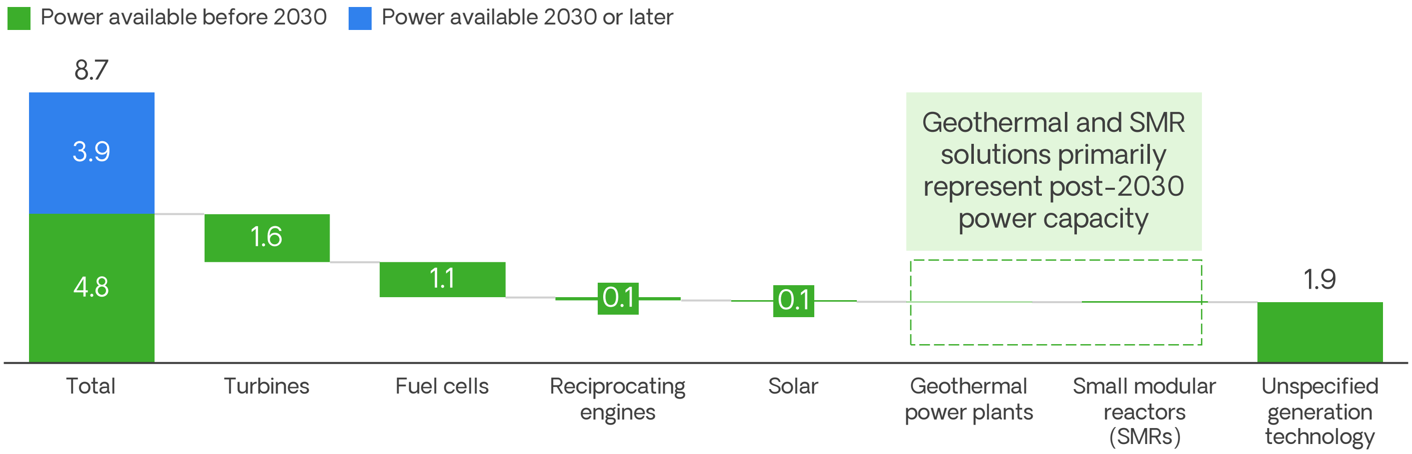
Emerging technologies

Data center leaders are showing growing optimism about emerging technologies such as geothermal power, small modular reactors (SMRs), and gas generation with carbon capture and sequestration (CCS). We are seeing these technologies appear in long-term roadmaps as companies work toward sustainability commitments.

Recent announcements related to these technologies focus on data centers expected to go online after 2030, due to the technologies' early stage of development. Natural gas solutions are expected to bridge the gap until net zero fuel options are ready for large-scale deployment.

Data centers are using several technologies to meet their onsite power needs

GW of announced onsite power generation for data centers[12]



Announcements of onsite power generation for data centers available before 2030 primarily involve **turbines, fuel cells, and reciprocating engines.** Other data centers are adopting emerging technologies such as **geothermal power and SMRs**, which we expect would largely become available after 2030.

Source: Publicly available data center announcements

12. Based on publicly available announcements - may not be comprehensive. If power availability timelines were not specified, natural gas-based power was assumed to be available before 2030, and SMR-based power was assumed to be available 2030 or later. For Meta's upcoming data center in Louisiana, 1GW of the 2GW total capacity was assumed to be available before 2030, while the other 1GW was assumed to be available 2030 or later.

4. Looking ahead, leaders expect a paradigm shift in 2025

These are challenging times for data centers looking to come online quickly. Yet there is also much excitement, with many data center leaders actively forging bold and creative partnerships with players across the power ecosystem. In 2025, we expect the continued growth of these three trends:

1. **Onsite power growth: More than 30% of data centers announced in the next 12 months will include some portion of onsite power.**

   Given the projected growth in data center power capacity and increasing expectations of onsite power adoption, we expect announcements to accelerate.

2. **Evolving buying criteria: Increasingly complex buying criteria will reshape the power technology market and establish new winners and losers.**

   Decision factors will go beyond traditional factors like reliability and cost. For instance, supply chain constraints for critical components should lead to greater adoption of technologies with quicker deployment times, such as fuel cells.

   Additionally, a small but growing number of data centers announced in 2025 are expected to run on "islanded power," operating independently of the grid. For AI data centers, the highly variable power loads will require technology that can combine energy generation with fast-response energy storage.

3. **Minimizing adverse impacts: Successful data center developments will use new technologies to minimize impact on local communities and the environment.**

   Leading organizations will actively collaborate with local communities to reduce adverse impacts from data centers. Decision makers will favor technologies with lesser permitting implications and greater environmental performance such as higher air quality or lower noise levels.

Next steps

Reach out to our onsite power experts
at bloomenergy.com/contact

Explore our solutions for data centers
at bloomenergy.com/industries/data-center-power

About Bloom Energy

Bloom Energy empowers businesses and communities to responsibly take charge of their energy. The company's leading solid oxide platform for distributed generation of electricity and hydrogen is changing the future of energy. Fortune 100 companies around the world turn to Bloom Energy as a trusted partner to deliver lower carbon energy today and a net-zero future.

Forward Looking Statements

This paper contains certain forward-looking statements, which are subject to the safe harbor provisions of the Private Securities Litigation Reform Act of 1995. Forward-looking statements generally relate to future events or our future financial or operating performance. In some cases, you can identify forward-looking statements because they contain words such as "anticipate," "believe," "could," "estimate," "expect," "intend," "may," "should," "will" and "would" or the negative of these words or similar terms or expressions that concern Bloom's expectations, strategy, priorities, plans, or intentions. These forward-looking statements include, but are not limited to, expectations for power demand growth and data center power growth to continue in the U.S., anticipated data center IT capacity announcements, percentage of data centers expected to employ onsite power generation, and expected growth in adoption of decarbonization options. Readers are cautioned that these forward-looking statements are only predictions and may differ materially from actual future events or results due to a variety of factors including, but not limited to, risks and uncertainties detailed in Bloom's SEC filings. More information on potential risks and uncertainties that may impact Bloom's business are set forth in Bloom's periodic reports filed with the SEC, including its Annual Report on Form 10-K for the year ended December 31, 2023, and Quarterly Reports on Form 10-Q for the quarters ended March 31, 2024, June 30, 2024, and September 30, 2024, filed with the SEC on February 15, 2024, May 9, 2024, August 8, 2024, and November 7, 2024 respectively, as well as subsequent reports filed with or furnished to the SEC. Bloom assumes no obligation to, and does not intend to, update any such forward-looking statements.

Bloomenergy®

bloomenergy.com